

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment for generating multiple possible solutions before selecting one. In this project, the team explored different ways to reduce metro ticketing delays, automate fare handling, improve ticket validation, and centralize booking information.

The shortlisted ideas were then grouped and prioritized using two major criteria: importance to the customer and feasibility of implementation within ServiceNow. The solution is built as a custom scoped application in ServiceNow Studio, so that all tables, catalog items, scripts and widgets are tracked together inside the application instead of the Global default update set.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Selected Problem Statement

Metro passengers need a faster and more reliable way to book tickets, know the correct fare, and receive verifiable digital tickets without depending entirely on manual counters or paper tickets. At the same time, metro operations teams need a centralized system that reduces manual intervention, avoids fare and data inconsistencies, and supports efficient ticket validation and request tracking.

Key brainstorming rules followed

- Focus first on the customer problem, not on a fixed technical solution.
- Generate several ideas without rejecting them during the initial brainstorming stage.
- Consider both passenger experience and metro operations requirements.
- Prefer solutions that can be built and maintained using ServiceNow low-code capabilities.
- Keep future scalability in mind, including payments, notifications, and analytics.

Step-2: Brainstorm, Idea Listing and Grouping

The team generated multiple ideas around the selected metro ticketing problem and grouped them by functional area so that overlapping ideas could be combined into stronger solution concepts.

Functional Area	Ideas Generated
Passenger Experience	• Self-service metro ticket booking through ServiceNow Service Portal • Simple Starting From and Going To station selection (reference variables pointing to the Metro Station's Details table) • Type of Journey (Single / Return), Number of Passengers and Payment Mode as guided form inputs • Request status tracking after ticket submission • Digital ticket instead of paper-based confirmation

Functional Area	Ideas Generated
Fare & Journey Logic	- Automatic fare calculation based on Starting From and Going To stations - Station-to-station fare mapping using station order and number of stops, grouped into fare bands (Rs. 20 / 30 / 40 / 50) - Automatic total based on Number of Passengers, with a 2x multiplier for return journeys - Amount fields (single journey / including return) auto-filled on screen and shown or hidden by UI policies depending on Type of Journey - Validation to prevent incorrect or incomplete route selections, including blocking the same station as source and destination - Future support for dynamic fare rules
Ticket Generation & Validation	- Generate a QR-based digital ticket - Show ticket details in a Service Portal-compatible modal - Use a widget-based approach (HTML template plus server script) for QR rendering - Allow station staff to verify ticket information quickly - Keep ticket data linked with the original booking request
Workflow Automation	- Use Flow Designer as the main automation engine - Create and populate metro ticket records automatically - Map catalog variables to the custom Metro Database table - Create tasks dynamically only when required - Use Flow Execution Details for troubleshooting
Security & Administration	- Role-based access for passengers, station staff, operations users, and admins - ACL configuration for ticket data protection - Keep custom scripting minimal for easier maintenance - Provide step-by-step setup documentation for another PDI - Maintain clear separation between requester and fulfiller activities - Build everything inside a custom scoped application for clean tracking and portability
Future Enhancements	- UPI / payment gateway integration (the catalog form currently captures the payment mode only) - Email-based digital ticket delivery - SMS / WhatsApp ticketing - Travel analytics dashboard - Passenger behavior and peak-time analysis - Lookup-table based fare rules for a larger station network

Grouped Solution Concept

The strongest combined concept was a ServiceNow-based digital metro ticketing system where a passenger books through a Service Catalog item, the system calculates the fare automatically on screen, Flow Designer processes and stores the booking, and a QR-based digital ticket is generated for validation. The design also supports role-based access, request tracking, maintainable low-code automation, and future integration with payment and communication channels.

Step-3: Idea Prioritization

Ideas were prioritized using two criteria from the template: how important the idea is to solving the customer problem and how feasible it is to implement within the available ServiceNow environment.

Idea	Customer Importance	Feasibility	Priority	Decision
Service Catalog ticket booking form	High	High	1	Implement
Automated fare calculation (client-side, station index based)	High	High	2	Implement
UI policies for conditional amount fields	Medium-High	High	3	Implement
Same source and destination validation	High	High	4	Implement

Idea	Customer Importance	Feasibility	Priority	Decision
Flow Designer automation	High	High	5	Implement
Custom Metro tables (Metro Station's Details and ticket records)	High	High	6	Implement
QR-based digital ticket (widget and server script)	High	Medium-High	7	Implement
Request status tracking	High	High	8	Implement
ACL and role-based access	High	High	9	Implement
Email / SMS ticket delivery	Medium-High	Medium	10	Future Enhancement
Payment gateway / UPI	High	Medium-Low	11	Future Enhancement
Travel analytics dashboard	Medium	Medium	12	Future Enhancement
WhatsApp-based booking	Medium	Low-Medium	13	Future Enhancement

Prioritization Matrix Interpretation

	Lower Feasibility	Higher Feasibility
Higher Importance	• Payment gateway / UPI • Email / SMS delivery	• Catalog booking • Fare calculation • UI policies • Same-station validation • Flow Designer • QR ticket • Custom tables • ACLs
Lower Importance	• WhatsApp booking	• Travel analytics dashboard

Final Prioritized Idea

Develop a ServiceNow-based Metro Ticket Generating System that provides a Service Catalog booking experience, automatic fare calculation with conditional amount display, Flow Designer-based automation, centralized station and ticket data storage in custom tables inside a scoped application, QR-based digital ticket generation, request tracking, and role-based access control. Payment gateways, SMS/email/WhatsApp ticket delivery, and travel analytics are retained as future enhancements because they add value but are not essential to the first working version.